Ops Field Guide - shipment delay alerts (Stage 1)

What this tool does: for each shipment snapshot it estimates the chance of ON_TIME, AT_RISK or CRITICAL_DELAY and raises an alert when CRITICAL is likely enough to be worth acting on. **It never reroutes, books or changes anything: every alert needs a planner's decision.**

The three statuses

Status	Meaning
ON_TIME	Expected delay under 6 hours [1]
AT_RISK	Expected delay 6 to 24 hours: watch it [1]
CRITICAL_DELAY	Expected delay of 24 hours or more: act [1]

When the tool raises an alert

- Alert when the chance of CRITICAL is **37% or more** (model `logreg`) [2].
- On held-out data it caught **79%** of critical delays; **38 of every 100 alerts** were false alarms [3].
- Why accept false alarms: a missed critical delay costs **10x** an unnecessary escalation (1 unit) [4].

Top five drivers of a CRITICAL prediction [5]

(Ranked by Role 1's SHAP analysis of the XGBoost model; the served model may weigh them slightly differently, and each API answer lists its own top drivers.)

1. `dwell_time_hours`
2. `capacity_utilization_pct`
3. `eng_labor_x_equipment_gap`
4. `weather_severity_index`
5. `roll_node_dwell_mean_24h`

In plain words (Role 2):

Logistics bottleneck risk is primarily driven by dwell time spikes, capacity utilization above 90%, and unexpected customs holds. When terminal dwell exceeds local baselines alongside high utilization, cargo faces compounded delays due to yard gridlock and crane shortages. Elevated historical delay rates and severe weather amplify this congestion by disrupting scheduled vessel berthing windows and feeder rail connections. To mitigate cascading delays, logistics operations teams must closely monitor seasonal demand surges during late summer and autumn, prioritizing expedited or cold-chain containers before yard saturation crosses critical thresholds. Automated predictive alerts provide actionable decision support, always requiring human operator review prior to enacting physical rerouting strategies. Continuous real-time tracking across international gateway ports ensures resilient end-to-end global supply chain transit performance.

What the tool will not do

- It does not execute reroutes, bookings or any shipment change.
- It is trained on calibrated simulation, not live port feeds; treat it as a second opinion.
- It is weakest at maritime chokepoints (Suez, Malacca, Panama): check those by hand [6].

Sources: [1] `contracts/label_policy.json` [2] `Stage01_ML/reports/eval_champion.json` [3] `Stage01_ML/reports/eval_metrics.json` [4] `contracts/cost_policy.json` [5] `Stage01_ML/predictions/feature_importance_matrix.csv` [6] `Stage01_ML/reports/eval_slices.csv`.

Generated by `python Stage01_ML/05_integration_engineer.py` on 2026-10-30.